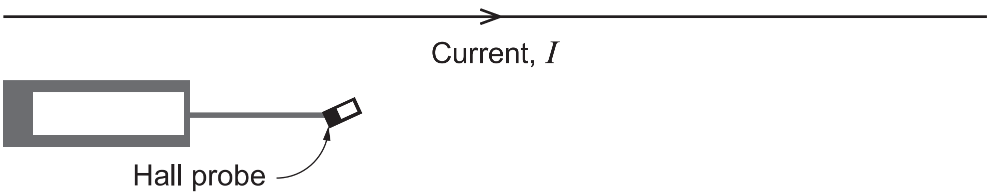
Examiner only

4. *(a)* The current in a long wire is 290 A and is too great to be measured by an ammeter. Explain how you could use a Hall probe, calibrated in tesla (T), to determine this current. [3]



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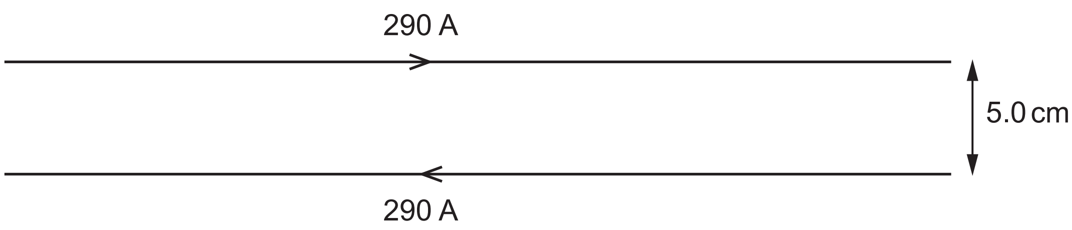
(b) Another long wire carrying the same large current is placed parallel to the original wire as shown. Calculate the force per unit length on each wire also stating the direction of the force on each wire. [4]



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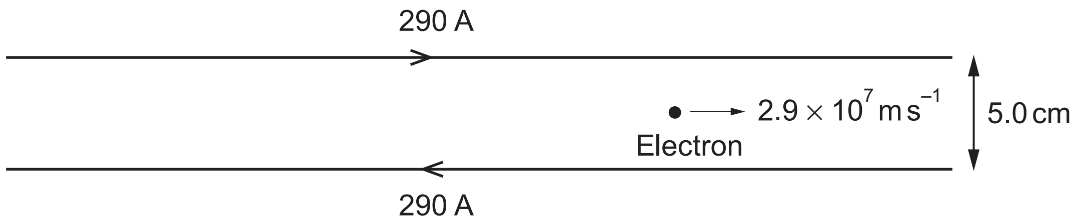
(c) (i) Tirion claims that an electron halfway between the wires, travelling at a speed of $2.9 \times 10^7\,\mathrm{m\,s^{-1}}$ parallel to the wires will perform perfect circular motion between the wires. Determine, using a suitable calculation, whether or not Tirion's claim is correct. The magnetic flux density halfway between the wires is 4.64 mT. [4]



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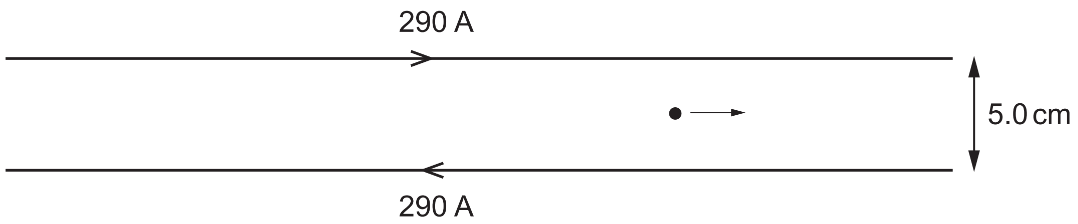
(ii) Sketch the motion of the electron. [2]



1420U401 13

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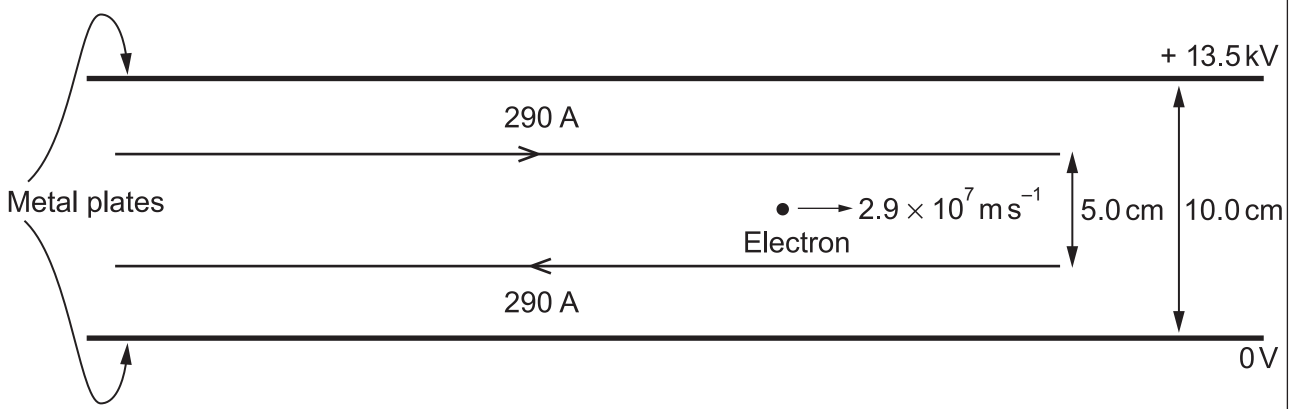
(d) Suppose that an electron travels in a region with a magnetic field and an electric field due to two parallel metal plates as shown below. Deduce whether or not the electron continues with constant velocity (B = 4.64 mT). [5]



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